

ActInf OrgStream #001.1 ~ Richard D. Bartlett

OrgStream | Richard D. Bartlett | 58:56

introducing myself

i can go on at great length i'll try to

be brief um i'm from new zealand that's

why my english is weird

i currently live in italy though um

because i wanted to be connected to

everything that's happening in europe um

i studied engineering at university so

i've got that frame of reference of

thinking in terms of systems and modular

parts and how things plug together

and in the last 10 years i kind of

drifted away from

technology and that kind of innovation

and got more interested in what you

might call social technology so how do

we organize people

what kind of systems support them to

work together effectively

but with a different set of values that

are usually applied in

most organizations so instead of for

instance prioritizing

efficiency or quality or

you know how quickly can we deliver

stuff i'm interested in some other

values which are more like

how do people develop as humans

how do we create relationships that

really

make a difference in our lives you know

how do we

pursue something that feels meaningful

and has a positive impact those kind of

values and then and then what kind of

organizational structures support those
kind of outcomes so that's my jam
i've done that in lots of different
formats so
building a startup company working in an
entrepreneurial ecosystem which is all
self-managed with a couple hundred
people
and now i do a lot of um
i guess designing
general purpose
community and organizational structures
that can be reproduced in lots of
different contexts
thanks for the intro and
i guess here on the slide we have one of
your ongoing
projects patterns for decentralized
organizing just
everywhere in there it's pretty cool
it's like pattern languages and
decentralization and organization all
topics that are important inside of
active inference as well as outside
and in this work as well as in a
episode 51 on the jim rutt show
you talked about this multi-scale view
on social systems
so we're going to kind of structure this
conversation along this ladder of nested
systems or nested processes we might
call them nested markov blankets in
active inference so maybe could you go
over this multi-scale nesting of social
structures from the individual on up and
why is it important to think about it
that way
sure

so

this is a lens that you can put on
to understand social physics
and it's not um
not making any claims about defining
reality or something like that but it is
a really useful lens to put on to like
what is going on in these group dynamics
in this organization in this community
so the smaller scale of the individual
and even at that scale the one person
it's useful sometimes to think of a
person as a group to think about
i'm not just richard i am
all of these different tendencies all
these different personalities
affordances
parts there's different parts of me some
some of my parts are
very enthusiastic and confident and bold
other parts are anxious and withdrawn
you know there's like all these
different tendencies going on
and
it's interesting and useful to think
about the metaphor between
like there's a kind of parallel between
the way that my parts are related to
each other
and the way that
i relate to other people
so there's like a kind of internal
relationship going on which is a
metaphor for the relationship that i
have with you
so if i am very
um
you know constricted and defensive

around if i want to disown my anxious
parts if i want to pretend like i don't
have anxiety
and just kind of like remove that whole
part
if you show up as my friend with anxiety
i'm going to be defensive against that
too it's like if i can't accept my
anxiety i'm not going to be able to
accept yours so this is a um a metaphor
that's been really fruitful for me
and
i i kind of came across it through some
amateur theorists and friends of mine
just thinking about this way and then
discovered that actually it's used in
um therapeutic context so there's like
transactional analysis internal family
systems probably a few other systems i
don't know so well therapeutic
modalities which which kind of treat the
individual as a group and use that as a
as a metaphor for getting insights into
what's going on in a person's life
so that's one that's one dimension
and and as i described that it sort of
implies
um
the way of working that i'm interested
in this more decentralized participatory
collaborative
shared power structures etc
it seems to involve or require
interior curiosity
that there's that that the
all of the agents have to be curious
about themselves and the impact that
they're having and what are their

motivations and values and drives and
why did i get really upset when that
happened and those kind of questions
sort of have to be on the table
so then the next scale is the dyad and a
dyad is just two people
and
at this point i think it's really useful
to pull in
rihanna eisler who's this amazing
theorist author historian
um and she has she's she's named this
framework which i find really helpful
which is the
partnership nomination spectrum
and it's a way of saying like
oversimplifying obviously all of history
and all of human relationships and
saying there's a spectrum in any
relationship which can be more like a
domination or more like a partnership so
more
partnership being we're different but
we're equal in some important way we've
got different competencies that we bring
together and a combination saying
there's like a ranking system where
one person has more power than the other
one person is kind of controlling the
other the other person is submitting
and in any interpersonal relationship we
can be
anywhere on that spectrum from very
powerish to very domineering
and not just at the interpersonal scale
but also at the very large scale so she
she in a lot of her work she's described
the connection between for example

what's happening in nazi germany
in the 30s and 40s
and what's happening in
german families at that time you know
how did those how do those two scales
interact with each other
um so the diet let's let's assume you
agree with my my values which is i'd
like to live in a culture that has less
domination less submission more
partnership
more equity
um the diet is the first place really
where we get to practice that
to say when i'm in a relationship with
you am i trying to assert dominance
or can i find a different way to relate
can i you know do we always have to
compete and make a ranking or can we
find different ways of linking and
making this like multi-dimensional
connections which are not so focused on
who's the best and the strongest and the
coolest and the fastest and the smartest
uh but just let's discover all of our
unique competences and see how they fit
together in interesting ways
so that's a diet scale and if you think
about
a really great friend
or a coach or a therapist or a partner
those are really crucial dyads
and
um
if you're interested in growing and
developing and maturing or evolving
i think a lot of that developmental work
happens in a dyadic relationship i think

there's often someone else there who's like you know sometimes we say holding the space you know it's a very vague term but they're they're creating some kind of nurturing container for you to do your thing

then the next scale up is called the crew and i use the word crew because i was thinking about a sailing ship you know it's a very active people have different roles

um

there's there's this expectation that we're different but we're also doing something together it's practical it's hands-on

and a crew is

small

i usually say it's the size of a dinner table conversation so like five people maybe and i think there's actually some good science on this now i was listening to a recent podcast with um

robin dunbar and he says that

five people

can have one conversation because they can all model each other's perspective simultaneously and kind of hold a shared state

and and it's kind of we can our brains can kind of natively manage that but six or seven people they just can't hold everyone all the different you know the combinations of that gets too too much complexity and so you'll notice this if you're having a conversation with five people and then two others show up that the conversation will split into

multiple sub-conversations just
completely natural happens all the time
so the point of focusing on the crew
scale is saying
there's something that we can do with a
group of four or five people
that we can't do with eleven
we can't do with seventeen
we can't do with two you know there's
unique affordances and competencies
for that group for that size and and a
big part of what i'm doing with this
framework that we call micro solidarity
is trying to train people in a literacy
of scales and understand
there's an articulation at each of these
different scales

um

the groups are good for different things
you know um if if you want to have an
intimate conversation probably a couple
of people but if you want to have like a
dance party you're going to need more
you know it's like we can do different
stuff with different groups

um

and

and the crew is like

when people are starting out in micro
solidarity practice the crew will often
be a space for

peer support belonging connection um you
know maybe you take a course together
and they're your buddies that you're
thinking through

this new information

um

it's it's kind of about connecting

and then once those connections are established then it might evolve to the next stage which is much more about action getting stuff done

um having some kind of shared investment collaborating on projects that really mean a lot to you starting companies uh that's that sort of activity

and so then the question is how do you find your crewmates because you don't have chemistry with everyone we don't we don't all fit together with everyone

and that's what what i call a congregation that's what that's for so the congregation is um

could be 30 people but i kept it around 150 or 200 people

so a group that's big enough to have some diversity

um but small enough that there's no strangers there that everyone has some sense of who everyone else is they might not be super close but we're on first name basis with everyone

and it's called a congregation because you congregate that's the main thing that happens is these um occasional gatherings where people come together and you get to meet all these other people and it's like a dating pool where you can find your crewmates

and this could be a conference it could be a retreat it could be you know a summer holiday there's lots of different ways of doing this

um

but again there's the scale factor where we say

there's a boundary here around 150 people maximum because again there's some things you can do with a group of that scale that you can't do with a group of 500 and especially if you're doing governance you can govern a group of 150 people basically with dialogue trust mutual exchange of like who is this person what do they care about like i can we can have good enough rough models of each other and then maybe articulate a few agreements or a few principles and that's kind of enough to keep the system coherent and organized you don't need a whole complex bureaucracy you don't need a lot of checks and balances you don't need some kind of legislation system or a police service or any of that sort of stuff which you do as you get into the larger scale so it's saying let's draw a boundary and say there's a kind of efficiency that happens at that scale and then in my framework at least for now anything bigger than 150 people is what i call a crowd because once you get beyond that scale you you can just assume that most people are going to be strangers to each other so you might for example go to a festival or a conference with 5 000 people and you might feel some sense of

imagined community some sense of like
these are my people
and that can actually be a little bit
confusing
because it feels like they're your
community
you don't actually have enough mutual
knowledge of each other that you can do
stuff you know so like if you
if you don't show up one day because
you're sick or there's some dramatic
thing happening in your family you can't
really count on those people to look out
for you because there's there's just a
lack of visibility you know there's just
too many too many relationships to track
and so i just put everything from 150 up
to 8 million in the box called crowd and
it's kind of like that's a that's a
problem for someone else to solve when
i'm focused on the small scale
thanks for the
overview and there's many
multi-scale systems and active inference
ideas that we'll
return to and i think how
those different levels of organizations
play out in the past present and future
of science and research and in other
domains i think we'll return to that but
let's go to relationship first so here
just to sort of bring the active
inference into play
is a figure from our
uh 2020 paper on
active inference and online teams and
online organization
so active inference as the name suggests

is about action and inference it's about
how perception
in the context of a generative model of
the individual
translates into action
and here we have two teammates so this
is really getting at that minimal
relationship the minimal dyad
and each teammate is perceiving
the digital information that's their
shared informational niche
and then they're each perceiving it
uniquely unpacking it in light of their
own experiences and then acting and
that's modifying the niche so this is
like two people on a video chat or
editing a shared document and this is
sort of an overview that is reminiscent
of many other perception action loops
although in active inference there's a
lot more
math and technical details that underlie
it
so
how do we design
for healthy interactions and multi-scale
resilience especially online knowing
that there's so many degrees of freedom
and so many ways to just each little it
point and interface in this diagram
how do we know how to design what works
this is so great i love looking at this
diagram and just like doing the
translation
between the cybernetics and and what i'm
into um
i mean what first comes to mind looking
at this diagram is the two the two

speech bubbles

um

as they come closer together

like in common parlance we call that

intimacy

i think like when we

when we get closer and closer in our

shared context

where i can trust that

i know you and you know me and that we

actually do have shared understanding

that um when i'm saying something you've

got more or less the same picture in

mind as what i've got that's intimacy

and that's um

this very

i mean the idea of like how do you

engineer intimacy is a very clumsy

concept you know it's very unfamiliar

concept but that's a lot of what we're

doing i think

um

another thing that comes to mind looking

at this diagram is

you've got the thought bubble of

perception

of course the perception includes the

other person and it also includes your

imagination of what the other person is

thinking about you

and and i mentioned interior curiosity

right this is all part of the whole

system of

um

when people are starting out in this way

of working in this

participatory

collaborative shared way of working

one of the things that often needs to be
developed is self-awareness
meaning
do i know what impact i'm having on the
group
so it's not just me broadcasting like
i've got these things to say and i just
show up it's also the kind of tracking
what's my impact
how do other people perceive me
um and and does my self-perception match
their perception and and there's always
a gap there what do i do about that gap
and how do i bring those two things uh
closer into alignment
and again in my experience that happens
in relationship that's not something you
can really do on your own
maybe some people know how to do it i
certainly don't know how to cultivate a
great deal of self-awareness on my own
it's always in these feedback groups
with other people that are that are
helping me to tune in my self-evaluation
um
so
there's some very concrete practical
stuff you can do to improve
self-awareness and relationships and
that can be like
um sometimes it almost sounds maybe
naive
or childish or something but
um
often we'll start with the very concrete
what we call listen and loop so
i'll share something and you'll reflect
back to me what you heard as close as

you can track it

and then maybe the next loop would be

um

i'll share something and then you'll

share what you heard was important to me

some kind of interpretation or some kind

of abstraction of what you heard and

we're and we're really as i hear what

you're saying back to me

um i realized ah that is what i said but

it's not what i meant and i need to add

this extra information and there's this

tuning process that's going on between

us

so we do a lot of that a lot of really

intentional like here's a process

do this then do this then you know

you're gonna have five minutes of this

person sharing and then three minutes of

this person repeating back and we kind

of drive people through these very um

defined choreographies and that

very quickly helps people to see

oh when i'm communicating people are not

tracking what i'm saying

you know

the thing i'm the reflection i'm getting

back doesn't actually feel very close to

the picture i had in my head

and it helps you

tune that and basically learn how to

communicate more effectively

so

in active inference we talk a lot about

having a shared generative model and

there's another figure where narrative

is playing the role of a shared

generative model and that concept of

thinking through other minds which is like our cognition is intrinsically about others world models as well as it including their model of us so these are all um important topics and things that again we kind of have the space at the high level mapped out the possibilities are there in the formalism and then which islands of parameter combinations actually lead to the kind of positive social relationships that we want to foster that's the empirical question and then also to connect that to the interior curiosity so the way that policy selection or basically behavioral selection works in active inference is it's always a combination of the pragmatic and the epistemic value so like what is the utility what is the external motivation of this action and then what information do i gain what is the knowledge gain um or the intrinsic motivation of the action and then just like you said with a choreography it's like contact improv that's at the physical level there's sequences of actions that dyads can participate in over the hour over the month over their lifetime so that they can learn how to have a better joint

model of inference and action so that
they can do the tango better or so that
they can do some sort of physical
operation better
choreography that we see in the
operating theater
and then we take it into the realm of
symbolic communication and so it's easy
to see when your hands don't match or
you look at the video you look at the
the golf swing but then in communication
it's a little bit of a less tangible
space and so it's really important to
have these kinds of practices that help
us calibrate our meaning transmission
i want to add on this you're getting my
brain going on so i just talked about
the diet scale

but obviously i'm always passing it
through other scales and seeing how do
we play there so at the crew scale um
what we discover is that in any group
you will have a unique contribution to
make to that group you'll be the best at
something you'll have you'll be the most
ready for some something and you don't
know what that is

people don't know what that is they
discover it through the reflections of
others so that's that's um

[Music]

that's a beautiful process actually but
in the larger scale when you get up to
sort of 30 people or 50 people
uh

there's something incredible that
happened to me like my whole process in
this accelerated hugely

through my involvement in
the occupy wall street movement where we
were doing a lot of consensus decision
making with 30 to 50 people sitting in
circles day after day after day of the
day

and what was happening there was um
you know we had a camp and we were
trying to organize our camp and people
needed a place to stay and to eat food
and then the media would show up and
have to talk to the media like there's
all these decisions that need to be made
as a community and we decided we're
going to do it by consensus which means
someone brings a problem or a proposal
or a need

and we're literally sitting around in a
circle

and it's as if as you hear from all the
different people you're just adding more
and more perspectives like these little
glimpses into truth that i can't see on
my own

but with the right kind of group process
and listening and like there's a ways of
assembling all of that knowledge into
like a three-dimensional picture
and actually it's like a 50-dimensional
picture when you've got 50 different
people with their own different models
and experiences and

and i got so motivated by that because
it was obvious that the collective
intelligence when the conditions are
right the collective intelligence is so
much superior to any individual
intelligence like

you the group with the right process the group is always going to be the smartest person in the room

and that is so motivating because once you see the the the brilliance of the intelligence and the creativity and the you know the joy of it it's there's so much enthusiasm that gets kicked off it's like oh i want to be part of this thing

and now me as an individual coming back to the individual scale

i often get feedback from people saying that like in my blog post for example that

i

that people value the way that i articulate complex things in simple ways the way that i can put into language experiences they've had but they've never quite known how to articulate and the reason i can do that is because of all of that consensus practice is because i've brought proposals and had them disagreed with and challenged from all these different perspectives and learned through many iterative loops how do i communicate in a way that many different people with all of their different ideological baggage are going to be able to listen and recognize the thing that i'm talking about

um and that's now one of my favorite skills that that that i get to bring to the group is like oh i know how to articulate stuff without alienating people along the way and so it's like we're going up and down the scale all

the time with this developmental process
awesome so you mentioned the person as a
group
and the the organism is involved in
unified action so if you know one leg's
walking backwards one's walking forward
just gonna rip in half so we're involved
in unified action that's when the
individual is acting as an individual
and consensus governance
is a protocol for generating unified
action
not necessarily unified inference
because you can have consensus it's like
yes i consent that our co-op will buy
almond milk
i have a different take on that than
somebody who needs to buy it for a
health reason but that's okay we have a
different unpacking we have a different
perception but we're involved in shared
action
and so that's a very
key point that
consensus allows us to have groups act
as individuals
but not necessarily in thought indeed
and that that may not scale to all
levels if one veto can take down a uh
a whole congregation's actions that's a
very different question so how do
governance mechanisms
change across scales is definitely
something you've um
brought up
let's go to this do you want to add
anything or we can go to yeah one thing
on this on this organism like this is

really animating me at the moment
um so in your body
it's clear like if i need to mute the
microphone because there's a noise going
on i just need to move one finger
right like that that's the
the action here
if i need to pick up my water bottle
that kind of takes my whole arm it's
going to do it's got to have this
coordinated action and let's say i need
then i need to go to the bathroom well
i'm going to have to move my entire body
and it's going to have to move as a
whole
and this is where consensus kind of
breaks down is because it usually treats
the group as a as a single limb
and what we need is the articulation
to be like there's some things that are
like go to the bathroom where you need
to move the whole group but there's lots
of things that are more like switching
off the microphone or taking a drink
where a much smaller subgroup is the
appropriate
site for the decision making to happen
and it's really difficult if you're used
to doing everything with the full group
to then break off into small groups
without it triggering a whole bunch of
conflict and alarm bells
and that's basically
what i'm trying to do like the very
biggest picture of what's going on in
the world how do we pull through with
climate change and inequality in society
and all these things that are going on

i think what we need is the ability to
articulate
different scales of coordination
something like climate change we need
the whole global body to move
and there's some things that are
happening in my neighborhood where we
just need one we just need the wrist and
the fingers you know
and we need to have in order to be able
to do that we need to be able to address
these different groups at different
scales
and and that means we need to have like
a nested collective identity that goes
all the way from
me and my family and my
flat and my building and my street you
know that kind of way wear them in
address uh it needs to go all the way up
to the global scale i need to be able to
sort of identify who am i
globally who am i in the in this bio
region and and
working at the small scale at least
let's track the self
the crew and the congregation there's
three scales at least can we get the
competency and the literacy there and
maybe that will equip us to be more
familiar and coherent and literate at
the larger scales that's kind of the big
project
okay very cool and just thinking about
like a active inference lab or the field
we have individuals who have buddies in
the lab we have crews that work on
specific projects

the organizational unit is more
verging towards the congregation there's
people who are on non-overlapping
projects and then a field is like a
crowd people check in and leave to
scientific fields all the time and
no one
knows or cares all the time so it's like
a music festival like you brought up
earlier so let's continue our walk up
the ladder although i suspect that we're
going to close the loop
so
um there's a few topics here and uh
dean
um and i spoke yesterday so it was a
good convo and we got a lot of these
things down so
first let's think about this crew and
congregation levels
this idea of a threshold model for
collective action
and the way that's related to skin in
the game so
previously i studied foraging regulation
in ants and we used threshold models for
ant decision making and they quite
literally have exoskeleton in the game
when they forage as nest mates and the
colony has
foragers on the line when they send them
out and then it's like
they take the model from neuroscience
with neural activation and we apply it
to the behavior of a different system
and then here we're thinking about
threshold behavior
in human systems and collective action

across scales so
where does threshold modeling come into
play for group decision making in people
so good no one's asked me that before
i hope i understand the question well
enough i'll try um
one thing that was really striking to me
in the first
so i'm in italy which is the first kind
of western country to have to deal with
the covert pandemic
and then
there was kind of maybe a month you know
times dilated but i think it was a month
or two where covert was very active in
europe but i hadn't arrived in the
states yet
and then there was this moment as it was
kind of people were realizing the
seriousness of it
and
people in the states weren't dealing
with it yet but they were awake to the
threat there was this this couple of
weeks a window of opportunity
where
i'm quite active on twitter i do a lot
of
connecting with people there and
thinking out loud there
and it felt like the entire network of
twitter was all focused on the one topic
of there's a pandemic coming
how do we get prepared
what's the truth
what do we know what do we don't know
all of this
and because everyone had skin in the

game

it was such a good conversation that i
was connecting into it was such high
quality

decision making and feedback and hey we
discovered this and this is a model that
you can use and did you realize that
this whole thing is corrupt and all of
this was happening very very rapidly and
very efficiently because everyone was
really they had skin in the game because
they were thinking about my family is it
is at risk here how do i keep them safe
it was very meaningful

to find this connection between the
global pandemic and the very local
immediate the choices that i make may be
the difference between life and death
for the people i care about
and that was the the most uh coherent
and high quality

exchange i've ever seen on twitter at
that scale

now most people don't experience twitter
that way most of the time

usually people experience twitter as a
much more polarized um you know there's
a lot of in-groups and art groups and
kind of battles around these political
ideologies and culture walls and so on
and i think a lot of that is about this
lack of skin in the game to a degree all
this confusion about skin in the game
where

there's a there's a mismatch between
my sphere of
like

influence and my sphere of awareness

so the last few decades of information technology have expanded my sphere of awareness to a global level where i can be aware of this horrible atrocity that just happened in this small country that i'd never heard of until yesterday um but my sphere of influence while it has grown like we've been we've got all of these extra capabilities because the internet it hasn't grown nearly as far as my sphere of awareness so it's like i'm continuously absorbed by all of the stuff that's happening and i've got all of this stimulation coming in but my impact is still relatively small it's much smaller than global i can't do anything about that atrocity that's happening in that small country i've never heard of before even though i can feel a lot of stress and like it's really important to me and i've got something to say on the topic and so that is part of what i think we're seeing happen on twitter is if someone is not really clear on their own agency like what impact can i have what decisions are up to me where's my autonomy how do i how do i achieve things in the world how do i do action that's effective not just as an individual but with a committed crew of people that are doing it with me or with a neighborhood or a community of people that are doing it with me if you don't have that

experience of agency and autonomy at those smaller scales

it's very easy to just

kind of get distracted by the global

scale and have an opinion about

everything and make everything into a

big drama and debate and organize

yourself into these polarized camps and

say i'm on the good team and everyone

else is a bad team it's really easy to

just kind of slip into that

um

and you're not achieving anything a lot

of the time you know it's like this

feeling of participation without

actually any impact

and once you get activated once you

discover your own agency once you

discover a few committed citizens who

are going to change the world with you

you know

once you have a

company or a crew or whatever then

suddenly you realize okay this is what i

can achieve like this is what's

realistic i can change the consumption

habits of my neighborhood that's

something i can do

i can't

change

the whole global economy that's not

really within my

sphere of influence and so i'm going to

spend more of my time focusing my

intelligence and my energy on the stuff

that i can make a difference on

and and that's another part of the

project here is to get people activated

into their agency and once they do that
they focus on the stuff where there's
skin in the game and and have a bit more
i don't know if it's like it's like
humility or modesty or something or a
little bit more of like
i don't actually know what to do with
the global scale
i'm going to relax a little bit on my
intense opinions about how i know what's
the right thing to do and just focus on
where i can have a meaningful impact
very cool so one key difference between
active inference and a lot of reward
centric frameworks is that in contrast
to
thinking that the maximization of reward
is the imperative for action
we frame the reduction of uncertainty as
the imperative for action so we can
update our models that's learning or we
can act to reduce our uncertainty and
that's what motivates behavior
and it has that pragmatic component
which is like utility a reward but it's
not just simply reinforcement or reward
learning
so we can connect that to what you were
saying there is so much uncertainty in
the world there's things we just simply
don't know about we don't know we don't
know about and now that
recently we have been able to have our
sphere of awareness expand to the global
or even the intergalactic scale
that breeds uncertainty of action
and so there is that overlap between our
sphere of influence and our sphere of

action

and we can think about how can we reduce
uncertainty if only to reduce our
anxiety or

feelings of negative valence but

actually

to be more effective as well locally

in our local situation we can reduce our

uncertainty about the um volatility of

the housing conditions locally and then

also um

that

aligns us with just

other people who we then actually in a

bottom-up way

start to be able to do more and more

people start connecting bringing their

perspectives and skills together

the awareness and the capacity for

action is increasing

because you entered at the level of

individual agency so that's like a very

important

step that when we

activate

we are also

starting to build from the bottom up

rather than just looking at how high the

mountaintop could be or looking at how

many mountain tops there are and there

must be infinite ways to get there as

well

i want to i want to rip on this

uncertainty thing that's really i've

never heard of that it's a really cool

way to think about it um so

we talked about how the sphere of

awareness is growing and we kind of have

been using these special metaphors
but it's not just that there's more
coming in it's also that there's more
incoherence so
we used to have this thing called the
six o'clock news and you would turn it
on and it would say these are the things
that happened today that are important
for you to know about
and everyone was referring back to the
same six o'clock news and then they'd go
to work in the next morning and then
have opinions they might disagree but
they were all
referring to a shared baseline
and the other thing that's happened with
the internet is that we don't have six
o'clock news we have all the time news
and it's coming from many different
perspectives and they all disagree with
each other and so we've lost that shared
baseline
now i don't think that six o'clock news
was ever telling the truth or was
particularly representative of reality
but it was simple it did provide at
least here's our zero line and we can
discuss here and there about it
um so there's so much more complexity of
perspective that we're having to
confront as well and so this the sense
of uncertainty is just magnified so much
not just because the volume of
information but the incoherence of it
and so how do we create what's the
shortcut to create certainty
well
there's two that i know very well which

you see online all the time one is moral
superiority like
me and my people are right
there we go solved all of the problems
the people that are like me are the ones
who are right all the other ones are
wrong they're ignorant they're stupid
they're evil like
that's bull
efficient and then the other one is
conspiracy theories where
um
me and my little cabal of people who
have somehow access to this superior
knowledge have
worked out there's a plot
and
if you just connect all the dots you'll
see how this is definitely what's
happening here's the plan this is who's
pulling the strings this is why you've
just got to follow the money man we've
got you know like that's really
flourishing at the moment as well
and it's this desperation to cling on to
a certainty
uh
and an incapacity for the degree of
uncertainty that actual reality actually
looks like when you start hearing from
all the different people simultaneously
so in active inference the preferences
are over sensory states so that's what
ties it to perceptual control theory
like we want to perceive temperatures
that we can live with it like
homeostasis
but to a large extent the inference is

on unseen causes of the world and so
there's interesting work about bayesian
views on conspiracy theory where you can
have a simple model that explains all of
the observables so that's um a nice
point and another way that we can sort
of shortcut the uncertainty is we can
make nested models like agree to
disagree so rather than uncertainty at
the sort of lateral level
you can encapsulate that in a model that
both people can actually have shared
precision on which is like we agree to
disagree or
um you know this is a really um wildly
inaccurate thermometer but we're gonna
kind of bound that within a way of
thinking about that thermometer so we're
not gonna be stressed out when the
thermometer goes up and down
we have a precise action
strategy for dealing with this
thermometer so then we start to identify
or seek precision at a higher level and
understanding that there's volatility at
a lower level
you wrote a really cool 2019 medium post
i will if you will
and
there's almost a few readings of even
the title
because
on one hand like i will do that if you
will do that but also to will is to want
so it's like i want to if you want to so
that actually speaks to how we want need
to have the shared action thresholding
but then there's that thinking through

other minds

shared intention

mapping

so

how does

i will if you will

scale or work in the dyad you know in

the i do's

that's the sort of minimal unit of i

will if you will

how do we

go to these levels like the crew the

congregation maybe even the crowd to

start thinking about i will if you will

bringing skin in the game and having

real effective action

i think this is legitimately one of the

the most promising places for

communication technology to make a big

difference on society and

for the good listen to have a positive

impact um and it's solving these kind of

coordination challenges where

um

one example that comes to mind i think

actually from that article

that kind of triggered this whole

research is it's quite common that that

tenants will have a bad relationship

with their landlord and the the landlord

has a lot of power and the tenants have

very little power and so they'll often

be in unfair conditions or they're being

mistreated or it's you know it's somehow

exploited of that situation

but because of the way um the

information is flowing in the way the

relationships are structured

there's there's kind of nothing that an individual tenant can do about it apart from finding another landlord often with the same problems

but if the tenants could be coordinated with each other then they can meet the uh

they can kind of counterbalance that that power dynamic

so

for example

i will if you will if you could get everyone in a building or in a neighborhood they had a shared like slumlord

landlord

to say i will go on rent strike if you do like if everyone in the block agrees we're not going to pay rent until we see this improvement in our housing then you better believe that they're going to get the action that they desire if they can coordinate that agreement and then say to the landlord look everyone

we've got this pledge here

you can't evict 300 people like just fix the leaky roof you know

um whereas if it's one individual that says to the landlord i'm not going to pay the rent well then you just get evicted

and so i think that example

applies to

how do we reduce carbon emissions you know how do we um confront the really large challenges we face in society often there's a disadvantage to being

the first mover or there's a disadvantage to doing something on your own without other people doing it with you and sometimes the disadvantage is so great that it's like it's meaningless i often say there's no such thing as a boycott of one you know i can i can decide to not fly but that doesn't have any impact on carbon emissions unless i'm participating in some kind of organized coordination movement where other people are also not flying with me and we're we're actively growing that shared power and we're saying look at all these people who've agreed not to fly we need to see electric airplanes please or more green forms of transit and and to be able to counterbalance that force um and so that's the kind of thing that can very easily be solved with the right the right precise use of technology which is um basically asking the right question of the right people and giving them the button to text that says yes if 10 000 people in my neighborhood agree to this action then certainly i would do it like i would love to do it but i'm not going to do it on my own and let's see if you can get that conditional commitment and if it reaches a certain threshold then yes we all act and that's the kind of problem that computers were basically

designed to solve i think so that gives
me a sense of
a little sense of optimism
yes um thinking about that flying
example it's almost like there's an
attractor and if you just pull out
individually you've reduced the demand
and actually that that might lower the
price so now another person jumps in so
as long as there's an attractor in the
system
then
uncoordinated action
is just
part of the
structure of that attractor
how do we actually shift the nature of
the attractor
that's where the coordination comes into
play and that's kind of what we
wrote down with like web 3 and
distributed autonomous organizations and
different kinds of cryptographic
technologies it's not just you're seeing
10 000 likes on something and that could
just be
basically a psyop on you from the side
of a server how could that be verified
information so you know those are real
people who have made a commitment to
action
that's when it starts to get very
interesting and then we wrote down a few
active inference terms like thinking
through other minds which we talked
about the importance of the counter
factual
because for any definite statements

there's so many counterfactuals we can go off into and when people are even talking about what kinds of worlds they want to see

it's sometimes a mixture of their preferences

as well as counterfactual action policies

so

it helps to have a formal model not the only possible way with active inference but a formal model so we can separate someone's preferences like what they do want to see

from counterfactuals on action

other things we could do

and then just separating the discussion

that way helps us be really creative

with the counterfactuals of action and

ask which will actually reduce our

uncertainty about realizing the

preference so um

that's hopefully going to be an active

and interesting area with getting people

to

um

structure commitment

because without a lot of intentionality

across different scales and careful

design the internet is a low trust place

so making it high trust at multiple

scales is very crucial

yeah the um the the piece about dao's

web three i guess that has some promise

i mean i should clarify i think the

concept of a dao de centralized

autonomous organization

from my research over the last couple of

months it means almost nothing like it doesn't really have a formal definition it's like a it's like a parenthesis that we put in the conversation which means all of my hopes and dreams about how organizations could be better which is a very useful object to have in a conversation but it doesn't tell you much about how to do it um

but i i'm very enthusiastic and optimistic about it because you've got these groups that are experimenting with resources that they really care about like a lot of the dows that are running have multi-million dollar treasuries that are being governed by tens up to thousands of people and they're confronting the real hard problems of how do you do that well one of the main bottlenecks is not like i described you know asking people the question and giving them the button to check it's the attention economy because to actually get people coordinated to look at hey here's the button you need to click that's something that you want to spend that wisely and then and spiral in the community that i've i've spent most of my time in we've sometimes even talked about that the attention economy is kind of like a shared credit card and you want to be really responsible when you spend that you can't be putting up a vote every

five minutes you've really got to be careful about that and that's something that um that's one example of a kind of bottleneck which requires a lot of sophistication and nuance and we don't really know how to program that yet that seems to be like a human strategic governance you know challenging communication problem totally agree about the sort of everything and nothingness of dao and i think maybe even the word dao is a subtle nod to that because distributed is on a continuum and there's not a single definition autonomous it's also on a continuum like people will say oh the embryo is a self-organizing system oh yes but of course there's the system that supports it so nothing is really self-arising so that's a continuum and kind of has nuance and then organization which is what people and other systems have been doing forever so it's kind of just saying we're using digital technology to organize which is awesome and it's many of those discussions that we would have in the co-op living room again about the almond milk but now it's happening with digital scarcity and real assets and lives on the line and then you you mentioned the attention economy so another active inference term is like regime of attention which is kind of just describing how attention and salience

is being distributed
through space and through time so just
doing a mindfulness exercise in the body
and okay pay attention to your feet and
your knee etc that is a
personal level regime of attention
and you're speaking to collective
regimes of attention
and then where do governance decisions
come into play are we going to
micromanage are we going to try to get
everyone's attention
or
might it be sort of a distributed regime
of attention that doesn't require as
much centralization
um
good
jumping off points for of course org
number one in our last few minutes
let's kind of
close with the crowd
and to uh
paraphrase slash mutate a quote
any sufficiently advanced crowd is
indistinguishable from an individual so
that's kind of where that multi-scale
ladder
loops back because of course we're an
ensemble of cells and so on so this
multi-scale nesting
it ends up kind of closing a loop
between a true crowd mob
and then the individual because once
it's a crowd it's just one you don't say
crowds it's just a singular so that
relates a lot to some technical
discussions we have about mean field

approximations like can you just assume
that that cup has infinite water
molecules just take the total average
or
how do we contrast that with a unique
trajectory based or a particle
particle-based approach so those are
physics questions but they come into
play with this dialectic between
individuals and crowds and then also
what's outside of the crowd or what is
the niche of the crowd so those were
some
crowd related thoughts we had any
anything to say on that yeah definitely
um probably the most inspiring examples
of
trying to advance participatory
democracy
um in the decade that i've been paying
attention to anyway they've been
happening in taiwan since 2014.
and one of the tools that they use is
called polis which sometimes they
describe as like a ai
ml facilitator
and
basically the way it works is you ask a
question of a large crowd
um or you you poll for opinions or
responses
and then
people can basically say i'm close to
this or i
i i agree or i disagree that's basically
the responses that they can do and then
it clusters all these sentiments and
says oh all of these people seem to more

or less agree with each other and these
ones agree with each other but there's a
disagreement between these two crowds
and so essentially what happens is you
take thousands of people
and you'll see there's like three or six
opinion groups
it's like it you can boil the crowd very
easily to a crew
and and then what you need to do well
first of all the crucial thing here is
it's a rough consensus it's not
consensus it's rough consensus it's like
there's there's a cluster here you know
like there's not a firm boundary but
there's a there's a kind of coherence at
that scale
and then once you've identified the four
or five opinion groups you just need to
find a trustworthy representative of
each one
bring them into a deliberation that's
facilitated by a human and broadcast
that conversation and and the five of
them sense making together and
uncovering okay what's our rough
consensus between the five of us and you
very rapidly generate very high quality
policy that people can get behind
to the extent that the quality of the
policy development they're doing in
taiwan
those kind of processes will get rubber
stamped by
the appropriate ministry and and
government to say yes we can we can
improve this right away so that's like
so inspiring to me and i guess the

process of going from those five opinion
groups to the one policy that everyone
can more or less live with that's kind
of the process of going from the crew to
the individual you know it's like this
is what the collective body can tolerate
or this is this is what we have in
common this is our island of agreement
so that's like
so incredibly exciting for me to to
track that happening
cool um i've heard of paulus and several
other
related tools in
metagov and we talk about these
different levels of governance that's
the body politic
and we can trust the process we can have
high confidence and precision on a
process
to really that highest level where you
just basically suggested that they have
so much trust in it that whatever the
outcome is whatever that thermometer
reads we trust the thermometer and so we
can have precision on that even if again
so trusting the process way of kind of
bounding our uncertainty
i think this is a fun
kind of closing notion for the
first time that we're speaking here
about theory and practice which is
something that came up in multiple um
podcasts and writings of yours
theory and practice
otherwise known as kind of inference and
action
they're inseparable

they're
the two swirls in that
tao
so
how do
theory and practice come together across
these scales
and just as we move forward how does our
individual theorizing and practicing
relate to these community levels
um great question the the
if you read the original micro
solidarity proposal
it's called a proposal because it starts
with this is my plan of action this is
what i'm doing
that's that for me the verb always comes
first
and then oh along the way i'll have to
explain some theory so you understand
what i'm talking about but it's always
for me it's really important to keep it
grounded in action i think there's um i
have a model in my head there's action
and then there's
talk about action and then there's talk
about talk
and i've got very little capacity for
talk about talk because occasionally you
need to go there but
i think it's very easy for a
a sort of culture to get
once one way or the other where it's all
action and there's no reflection there's
no thoughtfulness
or it's all talk about talk and there's
never really delivery there's never any
action so

it's really important to me to have a
balance there but a balance that's
biased towards the verbs

i'm on a war against nouns you might
actually subtly notice that in my
website that i try and get rid of the
nouns wherever i can

um

i think the other thing on that
is again a very practical
concrete step that we recommend in all
of the crews that we support to form

is to institute a rhythm of reflection
intentional reflection so it might be

one in five or one in ten of your
meetings should be spent looking
backwards retrospectively and saying
looking back on the last five meetings
we had

what was good that we wanted to do more
of

what was irritating let's do less of
that what are we going to do differently
in the next cycle

and cultivating that that shared
landscape of how do we view this shared
project that we're doing and what kind
of adjustments can we make and putting
that into a rhythmic context where it's
a dependable this is another way to
reduce uncertainty so we don't know
what's going to happen next but we do
know that every five meetings we have a
retrospective and that that rhythm is
dependable and that settles people
to make peace with all the change and
the uncertainty and that you know the
disruption that comes along it's one of

the things that you can really
count on and rest on so that's like
completely essential as a practice for
me and i'm always thank my theory i
think is always um
four or five steps behind my practice
i'm always just like let's just do the
work and
talk about it afterwards
well multi-scale nested rhythms happen
in the brain like the eeg rhythms as
well as a monthly annual etcetera
and
our
lab motto is act infer serve because we
have to act first that's that critical
action orientation
and when we act we start to reduce the
space of the counter factuals we're kind
of like plucking the strings that are
the sinews of the causal structure of
the world not that we see the strength
themselves but when we observe the
outcomes we start to learn more and so
when we have to act first though if we
start with inference if we start with
just theory
some people do exit that attractor but
also there's so much to think about and
we're never going to reduce our
uncertainty by thinking alone or we
could just reduce our uncertainty and be
super precise and incorrect
so that's like a very important call
if you have any final thoughts please
feel free to add them otherwise richard
this was an awesome first or extreme and
we really appreciate the visit thanks

daniel i yeah i could go on many
different tangents and rant at great
length about plucking the strings of
causality which kind of starts to touch
into my own homemade cosmology but i
think we'll save that we'll save that
for later the rant about the cosmic arts
accord

maybe i can come back for another
conversation sometime very cool so
you're always welcome and
i hope to continue the conversation with
you so again
great conversation and
good luck with everything you're working
on richard

thanks so much i would just say to the
lovely people listening if you if if you
like my way of thinking join me on
twitter that's where my stream of
consciousness is broadcasting into the
void 24 7. um and a more slowed down
from from my twitter page richard
you'll find like links and other things
if you want to go deeper into some other
projects and
follow websites and things like that
great very much recommended so see you
later bye